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Viscometric flow for a many-body dissipative particle dynamics (MDPD) fluid with Lees–Edwards boundary condition

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ABSTRACT

Viscometric properties of polymer are explored by the many-body dissipative particle dynamics (MDPD) using Lees–Edwards boundary conditions. The equation of state for the MDPD system is modified by fitting the density correction to different values of the cut-off radius. Due to the many-body interactions in MDPD, the viscosity contributed from the conservative force increases considerably with increasing repulsive coefficient, density and cut-off radius, and cannot be ignored compared to the ‘standard’ DPD case. The influence of these parameters on the MDPD viscosity is investigated, and we propose an equation to predict the viscosity in MDPD model. Additionally, the dumbbell polymer suspension model is investigated in the MDPD fluid, and the relations concerning first normal stress difference and shear rate, the relaxation time and spring constant, are consistent to theoretical works. We conclude that the MDPD model can be used to investigate the dynamics of non-Newtonian droplets.

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Many-body dissipative particle dynamics; viscosity measurement; polymer suspension; Lees–Edwards boundary condition

1. Introduction

In the past decades we have witnessed a tremendous growth in computational capabilities, which has contributed significantly to the development of numerical modelling and simulations. In particle-based type of techniques, the molecular dynamics (MD) technique is still a prevalent simulation technique at the molecular scale, but requires intensive computational demands due to the considerably small time and length scales. Meanwhile, complex hydrodynamics phenomena in mesoscale, such as many-body interactions in soft condensed matters and polymer suspensions, are also very topical. Relatively new simulation methods at the mesoscale, including the Lattice Boltzmann Method (LBM) [1,2], the Brownian Dynamics (BD) [3] and the Dissipative Particle Dynamics (DPD) methods, increasingly become important. The latter method (DPD) was firstly proposed by Hoogerbrugge and Koelman [4] and subsequently improved by other researchers [5–8]. DPD is conceptually considered as a combination of MD (continuous position and velocity), LBM (discrete time and spatial steps) and BD (Brownian stochastic force) [9]. Therefore, DPD has become versatile in modelling colloidal dispersions [10], red blood cells [11,12] and polymer solutions [13,14]. However, in the standard DPD, the weak conservative force results in an equation of state (EOS) which is quadratic in the density [15,16]. Consequently, the DPD model fails to accurately depict the fluid/fluid and the fluid/vapour phase transition [17]. In addition, the dimensionless ratio between isothermal compressibility and the pressure never

exceeds 2 irrespective of chosen parameters, whereas it should be of the order 10^4 for water under standard conditions [16]. Because of these disadvantages, as a possible remedy, a modified DPD model whose potential energy depends on the local density has been proposed and is referred to as the many-body dissipative particle dynamics (MDPD) [15–17]. In this MDPD model, an attractive force is also allowed, generating a surface tension, i.e. allowing the coexistence of vapour and fluid phase [18,19]. A diverse range of MDPD studies have been carried out, involving surface tension calculations [20,21], multi-phase flows [22] and droplet dynamics [23–25].

Generally, calculating viscometric properties (viscosity and normal stress differences) of the fluid is the first step in probing the hydrodynamics via DPD and MDPD methods [26]. In particular, the viscosity of a DPD system has been widely studied, both from computational and theoretical perspective. For instance, the viscosity of the fluid is studied by Meier et al. [27] through equilibrium method, i.e. Green–Kubo equation [28,29]. However, statistical issues [26], computing efficiency [30] and limiting to Newtonian fluids [31] pose a challenge for equilibrium numerical measurements. Thus, non-equilibrium methodologies have been recently utilised. For example, Backer and Fedosov [31,32] adopted the reverse Poiseuille flow (RPF) to calculate the viscosity of a polymer suspension in DPD flow and compared their results with previous literatures. The simple shear flow, with a constant shear rate, is arguably the most convenient method to study the viscometric flow of DPD fluids [33–35]. Meanwhile, large number of theoretical analyses,

based on Mori formalism [7,36], Fokker–Planck–Boltzmann equation [8,9] and Boltzmann pair collision theory [37,38], are available. Among these theoretical predictions, Marsh's kinetic results [9] are the most starting reference points in verifying DPD simulation results [10,26,31,39]. However, it should be especially reiterated that Marsh's results are for a DPD system without conservative force. Español [5] pointed out that the contribution to viscosity from the conservative forces could be merely neglected under high enough viscous damping parameter. Similarly, Jung et al. [40] pointed out that the kinetic contribution of viscosity is underestimated and the fluctuations in the pressure influence the viscosity as well. It is noted that Marsh, in his origin paper [9], stated that the potential energy contributions of viscosity were non-negligible for dense gases and liquids. In the MDPD system, the modification of the conservative force, to allow it to depend on the local fluid local density, leads to a considerably high pressure and density compared with the standard DPD fluid. To our knowledge, a systemic investigation of relation between the MDPD parameters and fluid viscosity is somewhat scarce, and the main aim of this work is to provide this investigation [41,42].

In this paper, we shed some insights on the effects of the MDPD parameters (attractive and repulsive coefficients in conservative force, density, cut-off radius, etc.) on the fluid viscosity, and provide a viscosity prediction for the MDPD system; the main properties of MDPD fluid can be obtained through initial parameters, and it is helpful in studying soft matter systems, multi-flow, etc. Furthermore, we demonstrate that a simple model of a dilute polymer solution can be accomplished in the MDPD framework, leading to the Oldroyd-B fluid. The paper is organised as follows: Section 2 revisits the theoretical background of MDPD method, the Lees–Edwards boundary conditions, and the elastic dumbbell model. In Section 3, we firstly match the equilibrium pressure in simulations with existing predicted equation, and modify this equation according to our simulation results. Next, the Lees–Edward boundary conditions are improved in order to generate a stable density, temperature, velocity profile and shear stress, irrespective of the dissipative parameter and shear rate. The effects of MDPD parameters on the fluid viscosity are then presented and a predictive equation for the viscosity is proposed. Lastly, a dilute suspension of dumbbells is introduced into the MDPD solvent; its first normal stress differences are verified with the theoretical result of the Oldroyd-B model. Finally, some concluding remarks are given in Section 4. The outcomes are helpful in any subsequent investigation of the dynamics of non-Newtonian droplets via the MDPD method.

2. Theoretical background

2.1. Many-body dissipative particle dynamics method

The conservative force in the standard DPD model is modified into the following form, depending on local density [15–19],

$$\mathbf{F}_{ij}^C = \left[A w_C(r_{ij}) + B(\bar{\rho}_i + \bar{\rho}_j) w_d(r_{ij}) \right] \mathbf{e}_{ij} \quad (1)$$

$$w_C(r_{ij}) = \begin{cases} \left(1 - \frac{r_{ij}}{r_c}\right) & r_{ij} \leq r_c, \\ 0 & r_{ij} > r_c, \end{cases} \quad (2)$$

$$w_d(r_{ij}) = \begin{cases} \left(1 - \frac{r_{ij}}{r_d}\right) & r_{ij} \leq r_d, \\ 0 & r_{ij} > r_d, \end{cases} \quad (3)$$

$$\bar{\rho}_i(r_{ij}) = \sum_{i \neq j} w_\rho(r_{ij}), \quad (4)$$

$$w_\rho(r_{ij}) = \frac{15}{2\pi r_d^3} \left(1 - \frac{r_{ij}}{r_d}\right)^2. \quad (5)$$

Here, $w_C(r_{ij})$ and $w_d(r_{ij})$ are two linear weight functions, vanish when the distance between two particles r_{ij} exceeds a cut-off radius, r_c or r_d , A and B are attractive and repulsive coefficient, respectively, controlling the interaction power between particle pairs. The instantaneous local density $\bar{\rho}_i(r_{ij})$ relies on the sum of density weight function $w_\rho(r_{ij})$, as shown in Equations (4) and (5), to create the van der Waals potential in the EOS, and $\mathbf{e}_{ij}$ is the unit vector, determining the direction from j -particle to i -particle.

The dissipative and random force, and the time evolution of each MDPD particle is analogous to DPD, following its Newton's second law as

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{F}_i = \sum_{j \neq i} \left(\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R \right). \quad (6)$$

The dissipative and random forces are considered as thermostat for DPD/MDPD model and obey a fluctuation–dissipation theorem [7]:

$$\frac{\sigma^2}{\gamma} = k_b T, \quad w^D(r_{ij}) = \left[w^R(r_{ij}) \right]^2 = \begin{cases} \left(1 - \frac{r_{ij}}{r_c}\right)^s & r_{ij} \leq r_c, \\ 0 & r_{ij} > r_c, \end{cases} \quad (7)$$

where σ and γ , $w^D(r_{ij})$ and $w^R(r_{ij})$ are the parameters and the weight function for the dissipative and the random force, respectively. Other thermostats, for examples, Lowe–Andersen (LA) [44] and Nosé–Hoover–Lowe–Andersen [45], are also possible candidates, but they are not considered here. Next, we establish the relationship between the physical units and the simulation parameters. Generally, there exist two methods [42,46,47] for this. Here, we adopt the parameterisation scheme according to coarse-graining level, presented by Ghoufi and Malfreyt [21,47]. The specific relations are listed in the Table 1.

2.2. Lees–Edward boundary conditions

The Lees–Edward boundary conditions were first introduced by Lees and Edward [48] and have been applied in MD to model a shear flow with a constant shear rate. The method is extended into DPD to eliminate the wall effect [43,49,50]. To realise the Lees–Edward boundary conditions, the periodic boundary condition is applied for the whole domain; on the top and bottom boundaries, meaning that when one particle leaves a boundary, it will be reintroduced into the opposite side with an additive velocity and a displacement. These values are given by

$$\Delta V_y = \pm \dot{\gamma}_z L_z, \quad (8)$$

$$\Delta D_y = \Delta V_y \delta t, \quad (9)$$

where L_z and $\dot{\gamma}_z$ are the length of the flow domain and the shear rate, respectively. After a simulation time, a stable linear velocity profile can be achieved. However, due to a relative velocity difference between the top and bottom boundaries, the density and velocity profiles may be unstable because of the velocity dependent dissipative force in DPD/MDPD system. Chatterjee [43] indicates that this problem is more obvious under a high dissipative parameter and it can be solved by ignoring the dissipative and random forces on the particles which cross the boundaries.

Table 1. r_c , r_d , ρ , δt , γ are the cut-off radius, density, time step and dissipative coefficient in MDPD unit, respectively. V and M are the volume (30 Å) and molar weight (18 g / mol) of a water molecule. N_a is the Avogadro's number, k_B is the Boltzmann constant and temperature T is fixed at 298 K.

MDPD symbol	MDPD units	MDPD units → Real units	Physical units
Beads	1.0	N_m	3H ₂ O
r_c	1.0	$(\rho N_m V)^{1/3}$	8.52 Å
r_d	0.75	$(\rho N_m V)^{1/3}$	4.83 Å
ρ	6.88	$\rho N_m M / N_a r_c^3$	997 kg / m ³
δt	0.01	$\delta t M^{0.5} r_c / (K_b \tilde{\eta})^{0.5}$	1 ps
γ	12.4	$\gamma k_B T / r_c^2$	70.6 mN/m

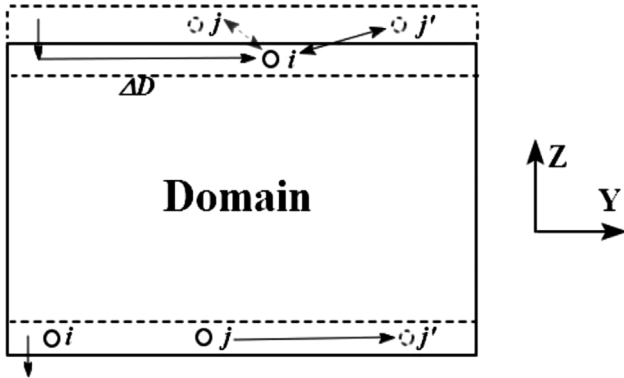


Figure 1. The schematic diagram of LE boundary condition in DPD. The particle i and j are undergoing LE boundary conditions; their interactions are calculated by particle i and j' , in which the fictitious particle j' merely undergoes periodic boundary condition.

Nevertheless, the dissipative and random forces act as thermostat in the DPD/MDPD model, and Chatterjee's solution [43] breaks the thermostat resulting in a loss of temperature control near the boundaries.

Hence, a remedy is adopted which is similar to Chen and Pan [49,50]. In Figure 1, when a particle i crosses the bottom boundary, it is applied LE boundary conditions by adding ΔV_y and ΔD_y . The interactions of particles are ordinarily calculated between particles i and j . However, the position and velocity of particle j is also artificial due to LE boundary conditions. So the abrupt variations of position and velocity between the particle pairs lead to incorrect fluctuations in velocity and density, which have been mentioned above. To overcome this, the correct interactions are supposed to be calculated between particle i and j' . The particle j' is merely undergoing periodic boundary (from the bottom layer to the top layer).

2.3. Elastic dumbbell polymer chain model

The elastic dumbbell polymer chain (or more general, the Rouse bead-spring model) is a simple model for dilute polymer solution [51]. In the context of MDPD, in this simplest dumbbell model, two MDPD particles are treated as two beads with a spring force between them. The spring force can be described as

$$F_{ij}^s = -H r_{ij}, \quad (10)$$

where F_{ij}^s is the connector force between particle i and j , H is the spring constant, and r_{ij} is an end-to-end vector between the two beads. In the MDPD simulation, the connector force is treated as an additional external force in the interactions. Note that the connectors, and the beads, are not physical entities, thus allowing them to cross each other: the chains are sometimes called 'phantom' chain [51]. This simple microstructure results in the Oldroyd-B model [51], in which the total stress, expressed as a sum of a solvent-contributed stress S^S , and a polymer-contributed stress S^P ,

$$T = S^S + S^P = 2\eta_s D + S^P, \quad (11)$$

where D is the strain rate tensor. The polymer-contributed stress obeys the upper-convected Maxwell model

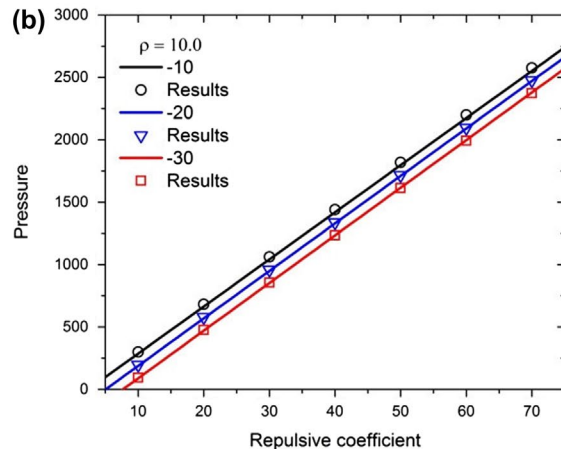
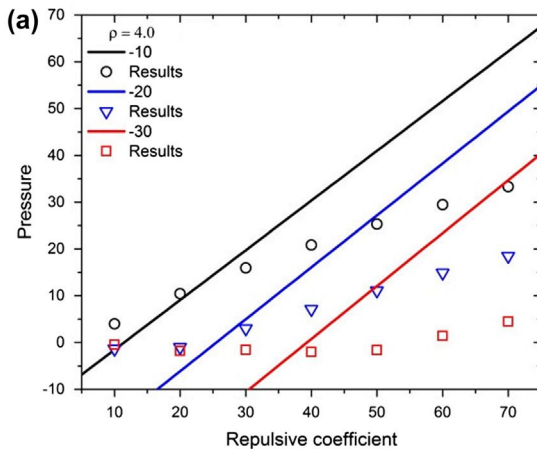


Figure 2. (Colour online) The numerical results for the pressure are matched to Equation (18) for different density, (a) $\rho = 4.0$, (b) $\rho = 10.0$ with fixed attractive and repulsive coefficient and $r_d = 0.75$.

$$\mathbf{S}^p + \lambda \frac{\delta}{\delta t} \mathbf{S}^p = 2\eta_p \mathbf{D}. \quad (12)$$

Combining Equations (11) and (12), we find

$$\mathbf{T} + \lambda_1 \frac{\delta}{\delta t} \mathbf{T} = 2\eta \left(\mathbf{D} + \lambda_2 \frac{\delta}{\delta t} \mathbf{D} \right), \quad (13)$$

where $\lambda_1 = \lambda$ is relaxation time, and $\lambda_2 = \frac{\lambda \eta_s}{\eta}$ is the retardation time, respectively. $\eta = \eta_s + \eta_p$ is the fluid viscosity, the sum of viscosity of solvent and polymer. Additionally, $\frac{\delta}{\delta t}$ is upper-convected time derivative defined as

$$\frac{\delta}{\delta t} \mathbf{T} = \frac{\partial}{\partial t} \mathbf{T} - \mathbf{L} \mathbf{T} - \mathbf{T} \mathbf{L}^T. \quad (14)$$

where $\mathbf{L} = (\nabla \mathbf{u})^T$ is the velocity gradient tensor, and the super-script T denotes the transpose operation. In a shear flow with a constant shear rate, this model yields a constant viscosity, a quadratic (in the shear rate) first normal stress difference and a zero second normal stress difference,

$$\eta = \frac{S_{xy}}{\dot{\gamma}} = \eta_s + \eta_p$$

$$N_1 = S_{xx}^p - S_{yy}^p = 2\eta_p \lambda \dot{\gamma}^2 \quad (15)$$

$$N_2 = S_{yy}^p - S_{zz}^p = 0$$

According to Equation (15), the relaxation time λ can be obtained by $N_1/2\eta_p \dot{\gamma}^2$. Furthermore, it is given by

$$\lambda = \frac{\xi}{4H}, \quad (16)$$

where ξ is the frictional (Stokes' drag) coefficient of a bead.

In the following section, the Lees-Edward boundary conditions are implemented to establish a simple shear flow for a dilute polymer solution, and we verify the MDPD simulation results to theoretical predictions.

3. Results and discussions

3.1. Validation of EOS for MDPD

In order to determine properties of MDPD model and link with physical parameters, the first step is to establish the EOS. The pressure can be measured in many ways, such as Kirkwood-Buff relation [52], the Irving-Kirkwood equation [53] or the test area method [54]. Here we choose the Irving-Kirkwood equation to calculate virial pressure,

$$P = \rho k_b T + \frac{1}{3V} \sum_{i=1}^{N-1} \sum_{j>i}^N \mathbf{r}_{ij} \cdot \mathbf{F}_{ij}^C. \quad (17)$$

A simulation domain with simple periodic boundary condition is adopted, and 4000 MDPD particles are employed. The density per unit volume varies from 4 to 10 via changing the size of domain. All simulation cases are fixed with $k_b T = 1.0$, $\gamma = 4.5$,

and $\Delta t = 0.01$. In addition, the system is allowed to equilibrate after 2×10^5 time steps to wash out transients and every sample is collected per 2×10^5 time steps. Taking sampling layers with same thickness along Z direction, the results are averaged in these sampling layers, and the percentage error is basically less than 3%.

In the previous work, Jamali et al. [41] proposed an EOS with respect to the attractive coefficient A , the repulsive coefficient B , the fluid density ρ and the cut-off radius r_d respectively

$$P = \rho K_b T + \alpha A \rho^2 + 2\alpha B r_d^4 (\rho^3 - c \rho^2 + d \rho) - \frac{\alpha B r_d^4}{|A|^{0.5}} \rho^2, \quad (18)$$

where $\alpha = 0.101$, $c = 4.69$, $d = 7.55$ are fitting parameters. This equation depicts the many-body interactions deriving from the density-dependent conservative force, and adds a new term which takes the combined influence of A and B into consideration. Based on Jamali et al.'s [41] descriptions, Equation (18) suggests that, at high density, the third order term in the density dominates the pressure. Firstly, we focus on matching our numerical results with this prediction.

Figure 2 shows the comparison between our numerical results and Equation (18). It can be seen that our results are consistent with this prediction at high density (Figure 2(b)). But at low density, Equation (18) apparently overestimates the pressure (Figure 2(a)), since the many-body interaction is not strong enough. From Figure 3, the influence of density on the pressure is illustrated. Equation (18) is able to accurately describe the pressure when the density exceeds 6.0, where the third order density dominates.

It is seen from Figure 4(a) that the numerical pressure is considerably larger than the suggested correlation Equation (18) at high values of the cut-off radius, even at a low density. It appears that Equation (18) provides only a qualitative fit to the data, shown in Figures 4 and 5. Jamali et al. [41] only considered a cut-off radius $r_d = 0.75$, where the quality of fit is fairly good. However, they did not realise that Equation (18) may not perform well at other values of r_d . High values of r_d will similarly strengthen the many-body interactions. Therefore, it is necessary to take cut-off radius into account and modify the

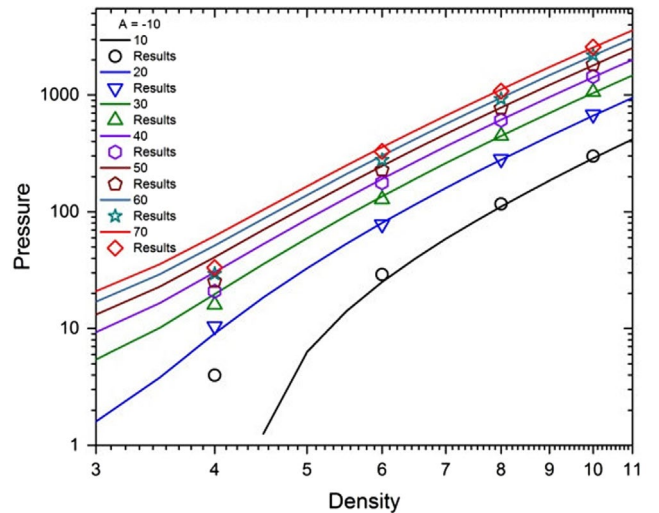


Figure 3. (Colour online) Relation between the pressure and the density, $\rho = 4.0, 6.0, 8.0, 10.0$, with $A = -10$ and $r_d = 0.75$.

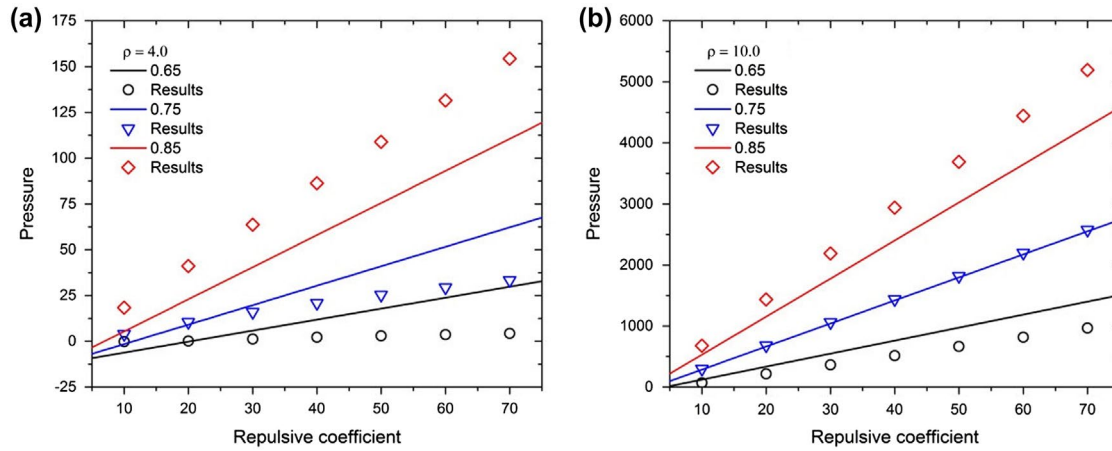


Figure 4. (Colour online) The numerical results for the pressure are matched to Equation (18) for different density, (a) $\rho = 4.0$, (b) $\rho = 10.0$ with given $A = -10$ and r_d varies from 0.65 to 0.85.

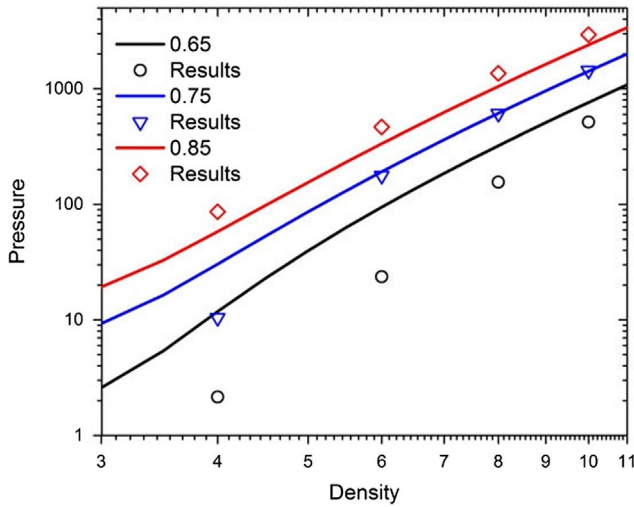


Figure 5. (Colour online) Relation between the pressure and the density, $\rho = 4.0$, 6.0, 8.0, 10.0, with $A = -10$ and $B = 40$ and r_d ranges from 0.65 to 0.85.

Table 2. Density correction c for many-body interaction is as a function of cut-off radius r_d .

r_d	0.65	0.70	0.75	0.80	0.85
c	6.4	5.43	4.63	3.92	3.44

Equation (18) so that it is capable of predicting the pressure under different r_d . To find a modified version of Equation (18), we revisit the original paper of Warren [19] and realise that the fitting number c , an empirical correction to the density in the many-body term, is supposed to depend on the cut-off radius r_d . Following Warren's suggestion, we calculate the specific coefficient c for different r_d , and the results are given in Table 2 and Figure 6.

With the new correlation for c , the quantitative fit between the numerical data and Equation (18) is excellent, as shown in Figure 7, irrespectively of the cut-off radius r_d . With the EOS for the MDPD model sorted out, we continue on investigation of MDPD viscometric.

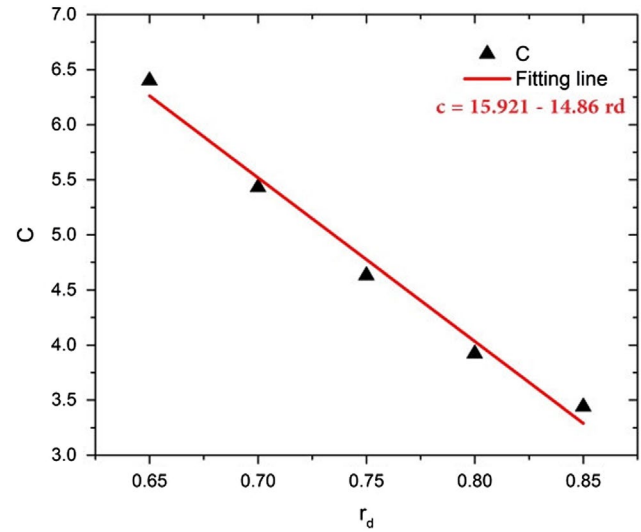


Figure 6. (Colour online) Density correction coefficient c is as a function of the cut-off radius r_d . The linear fitting equation suggests that $c = 15.92 - 14.86 r_d$.

3.2. Validation of Lees–Edward boundary condition for MDPD

Lees–Edward boundary conditions are a convenient method to generate a simple shear flow. It also allows the rheology of non-Newtonian suspensions to be studied, offering an alternative to the RPF method [26,31]. However, the interactions of particles near boundaries result in a non-linear and oscillatory velocity profile there at a high dissipative parameter γ . Chatterjee [43] ignores the dissipative and random forces between particles near the boundaries and obtains a stable velocity profile. Other researchers apply different thermostats [34,35] to stabilise the temperature. However, these thermostats may influence the transport properties of DPD/MDPD models [26]. Moreover, the shear flow creating by moving walls has its own challenges, such as obvious fluctuations near to the walls, violation of momentum conservation, unsolved coarse-graining level [26]. We still prefer the Lees–Edward boundary conditions

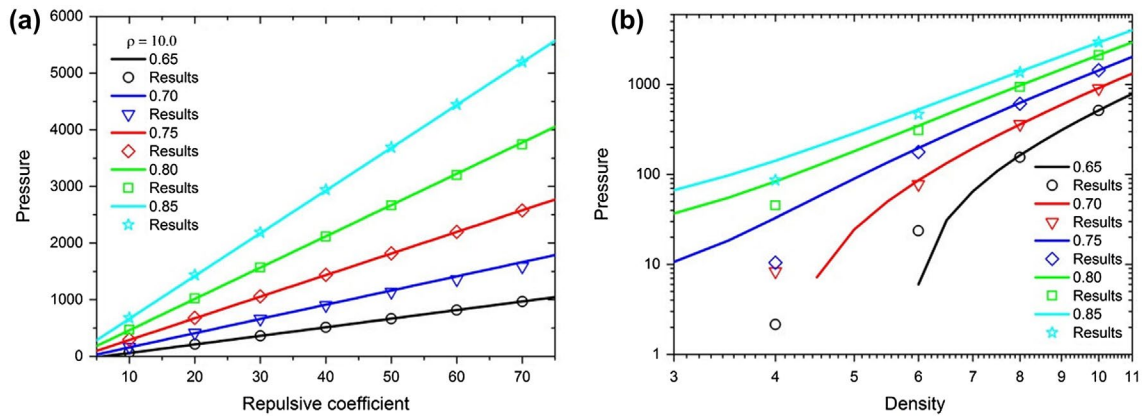


Figure 7. (Colour online) The numerical results for the pressure with modified EOS equation. (a) At different B values for $\rho = 10.0$, $A = -10$ and r_d ranged from 0.65 to 0.85. (b) At different ρ values for $A = -10$, $B = 40$ and r_d ranges from 0.65 to 0.85.

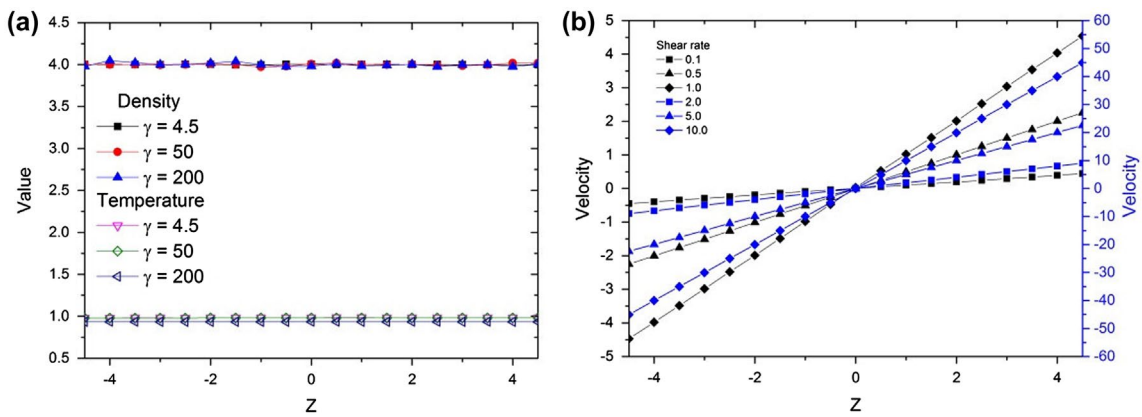


Figure 8. (Colour online) (a) The density and temperature along Z direction with $\gamma = 4.5, 50, 200$. (b) The velocity profile along Z direction with shear rate $\dot{\gamma}_z$ from 0.1 to 10.0.

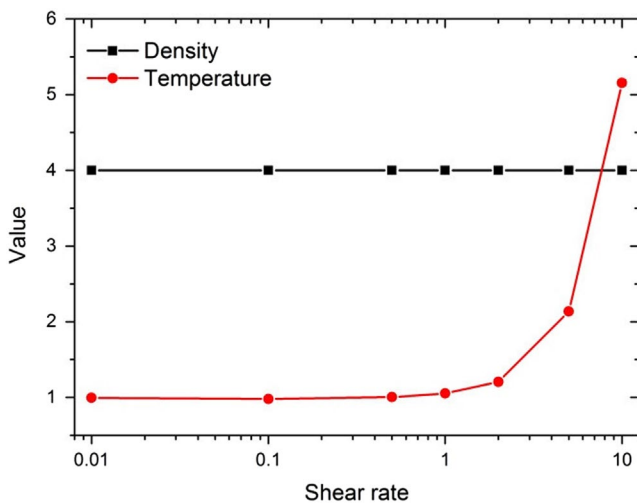


Figure 9. (Colour online) The variation of density and temperature with increasing shear rate.

with fluctuation-dissipative. The details are already expressed in Section 2. This procedure is validated by varying the dissipative parameter γ , as shown in Figure 8.

Figure 8(a) shows a stable density and temperature with a large range of the dissipative parameter γ via time step $\Delta t = 0.01-0.001$

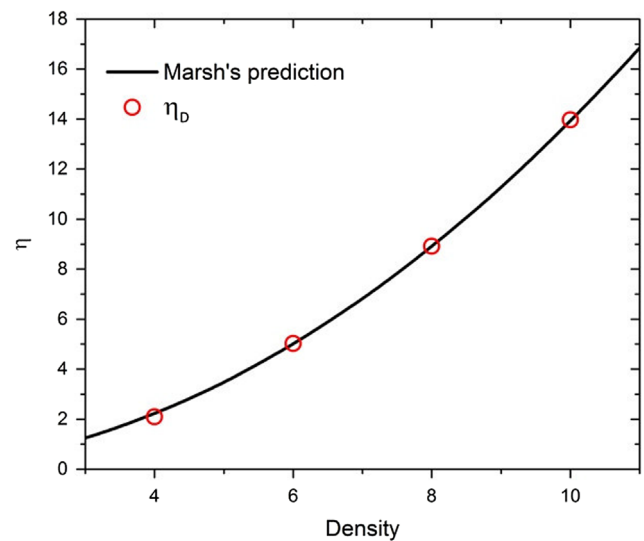


Figure 10. (Colour online) Matching η_D with Marsh's prediction [9], $\rho = 4.0, 6.0, 8.0, 10.0$. $\eta_D = \frac{512\pi\gamma\rho^2r_d^5}{51,975}$, where an exponent $s = 1/2$ in Equation (7) for the weight function of dissipative force is used [59,60].

(lower Δt at higher shear rate). The temperature is somewhat lower (by approximately 6%) than the initial temperature of 1.0 at a considerably high value of $\gamma = 200$. Additionally, the velocity

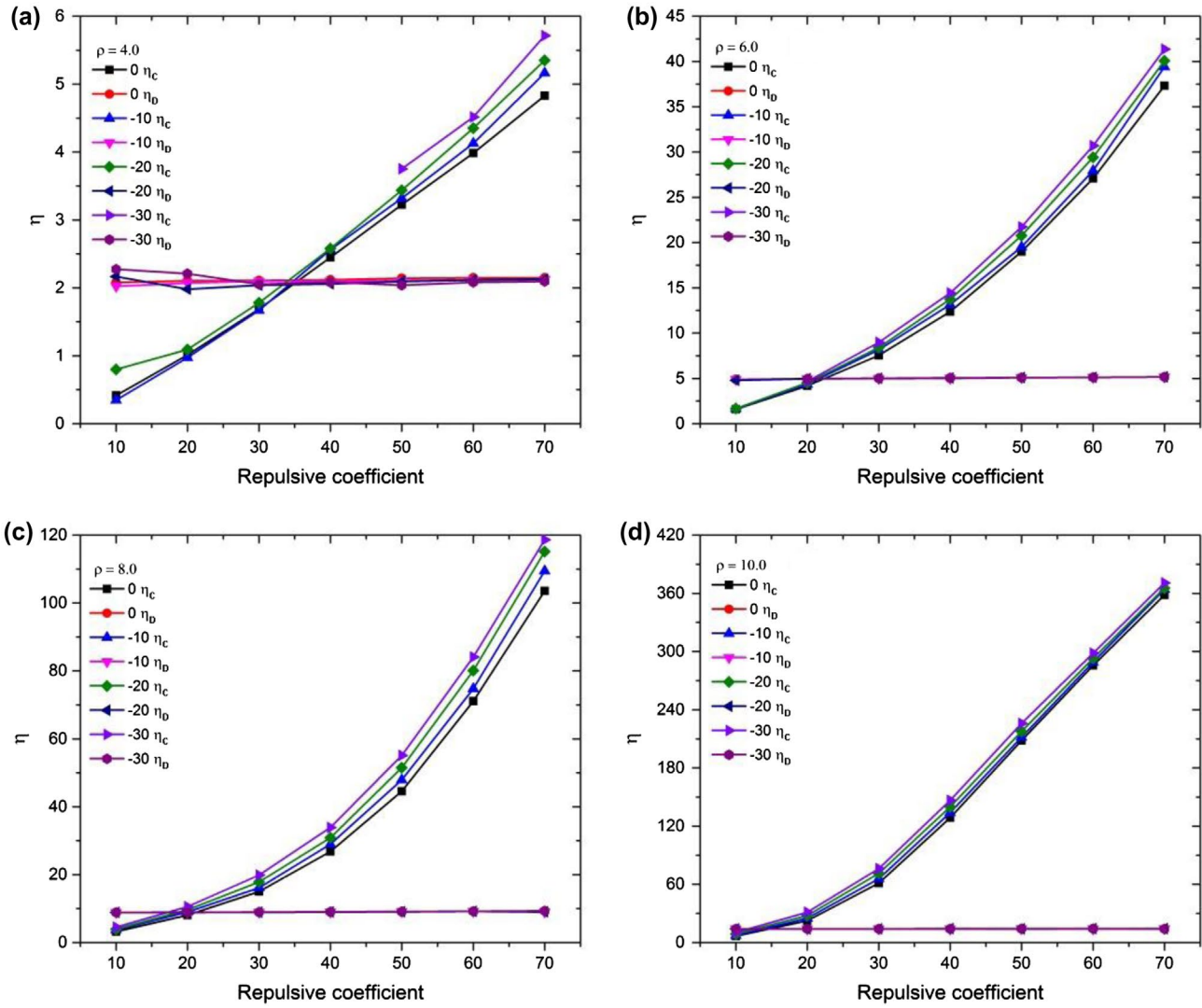


Figure 11. (Colour online) The conservative viscosity η_c and dissipative viscosity η_D with increasing attractive and repulsive coefficient at $r_d = 0.75$, for (a) $\rho = 4.0$, (b) $\rho = 6.0$, (c) $\rho = 8.0$, (d) $\rho = 10.0$.

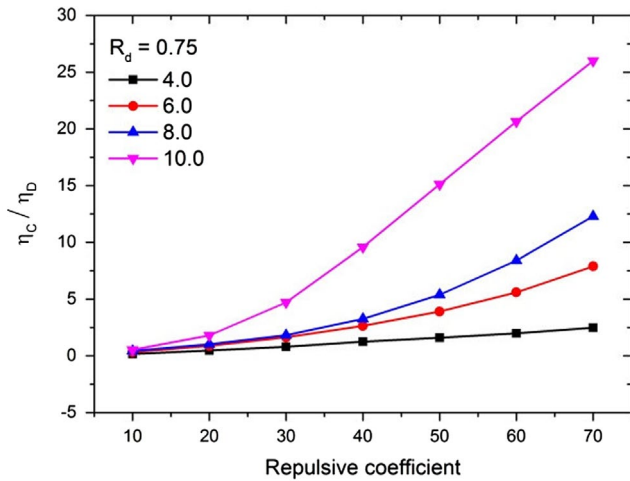


Figure 12. (Colour online) Ratio of conservative viscosity η_c to dissipative viscosity η_D with increasing repulsive coefficient at $A = -10$ and $r_d = 0.75$, for $\rho = 4.0, 6.0, 8.0, 10.0$.

Table 3. The correction parameters k and a in Equation (20), as a function of r_d , for predicting conservative viscosity in MDPD system.

r_d	0.65	0.75	0.85
k	0.46×10^{-6}	2.64×10^{-6}	11.5×10^{-6}
a	4.8	4.4	4.0

profile is also stable and linear regardless of shear rate. Next, the appropriate range of suitable shear rates need to be determined. Because under a high shear rate, unphysical shear-thickening behaviour, sharp increase in temperature and a deviation from linearity of the velocity profile have been observed [10,26,55]. Figure 9 shows the temperature and density with shear rate, indicating that a suitable range of shear rate should be less than 1. Even a small time step $\Delta t = 0.001$ cannot keep temperature stable [26]. Therefore, for convenience, we choose the shear rate at 0.1, dissipative parameter $\gamma = 4.5$ and initial temperature $k_B T = 1.0$ for our simulations, under $\Delta t = 0.01$.

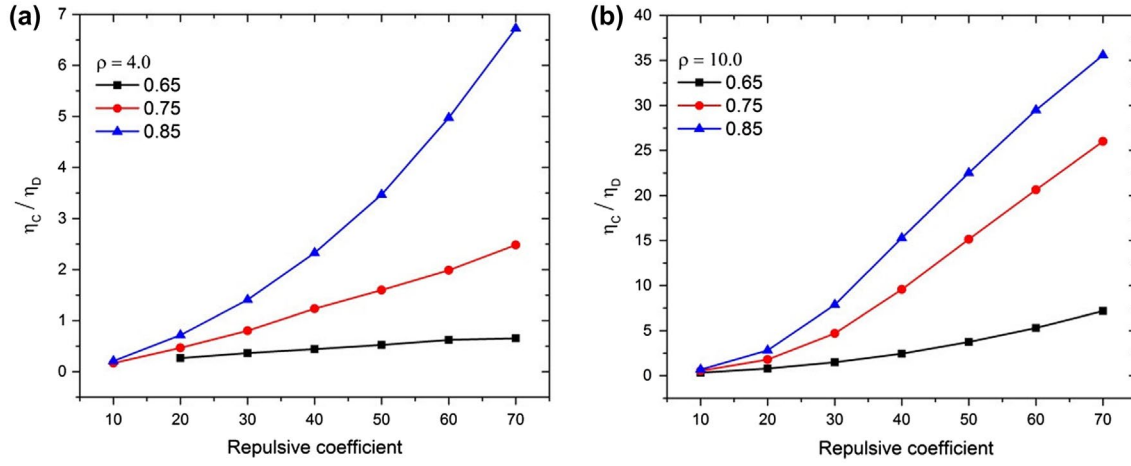


Figure 13. (Colour online) Ratio of conservative viscosity η_c to dissipative viscosity η_D with increasing repulsive coefficient at $A = -10$ and $r_d = 0.65, 0.75$, and 0.85 under (a) $\rho = 4.0$, (b) $\rho = 10.0$.

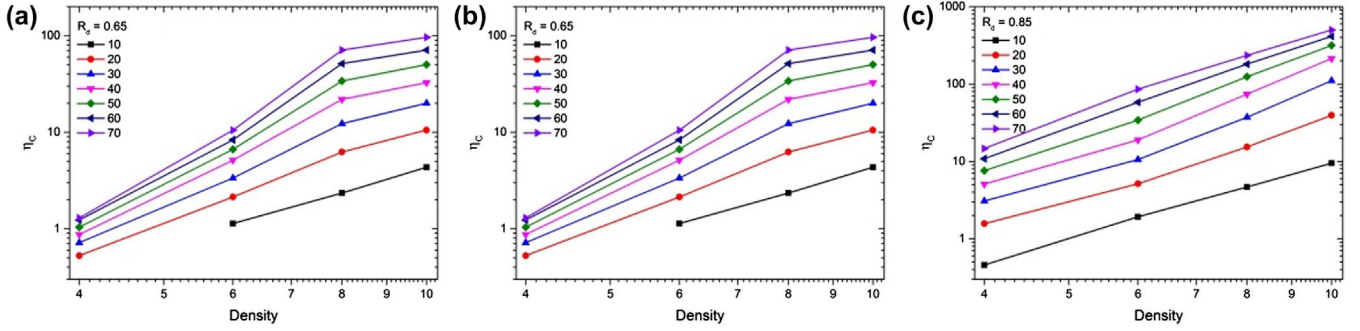


Figure 14. (Colour online) Logarithm diagram of η_c and ρ with $A = -10$ and increasing repulsive coefficient under (a) $r_d = 0.65$, (b) $r_d = 0.75$, (c) $r_d = 0.85$.

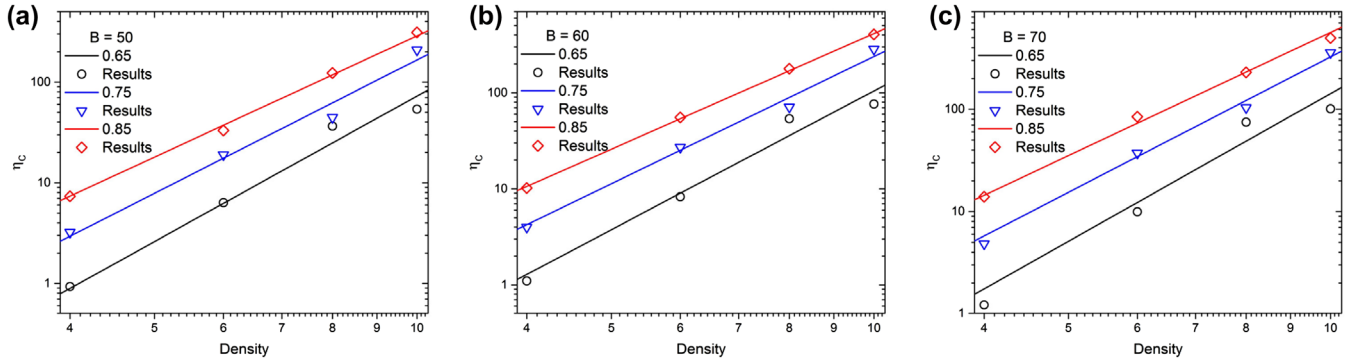


Figure 15. (Colour online) The numerical results for the conservative viscosity are fitted to Equation (20) for various ρ under different repulsive coefficients, (a) $B = 50$, (b) $B = 60$, (c) $B = 70$.

3.3. Viscosity measurement

Now, we focus on the viscosity of a MDPD fluid. The theoretical prediction by Marsh [9] is confined mainly to ideal DPD gases and has been much discussed [8,56,57]. The viscosity of DPD/MDPD consists of three parts, $\eta = \eta_K + \eta_C + \eta_D$, representing kinetic, conservative and dissipative viscosity, respectively [58]. The contributions, η_K and η_C , are neglected only under a high enough dissipative parameter. In our simulation, the kinetic contribution is ignored, because the collision time t_w is significantly larger than the kinetic relaxation time t_ρ , due to the relatively

low temperature, i.e. $k_B T = 1.0$. Therefore, conservative and dissipative interactions need only be taken into consideration. This is done by separating the conservative and dissipative forces in their contributions to the stress,

$$S_C = \frac{1}{3V} \sum_{i=1}^{N-1} \sum_{j>i}^N \mathbf{r}_{ij} \cdot \mathbf{F}_{ij}^C,$$

$$S_D = \frac{1}{3V} \sum_{i=1}^{N-1} \sum_{j>i}^N \mathbf{r}_{ij} \cdot \mathbf{F}_{ij}^D. \quad (19)$$

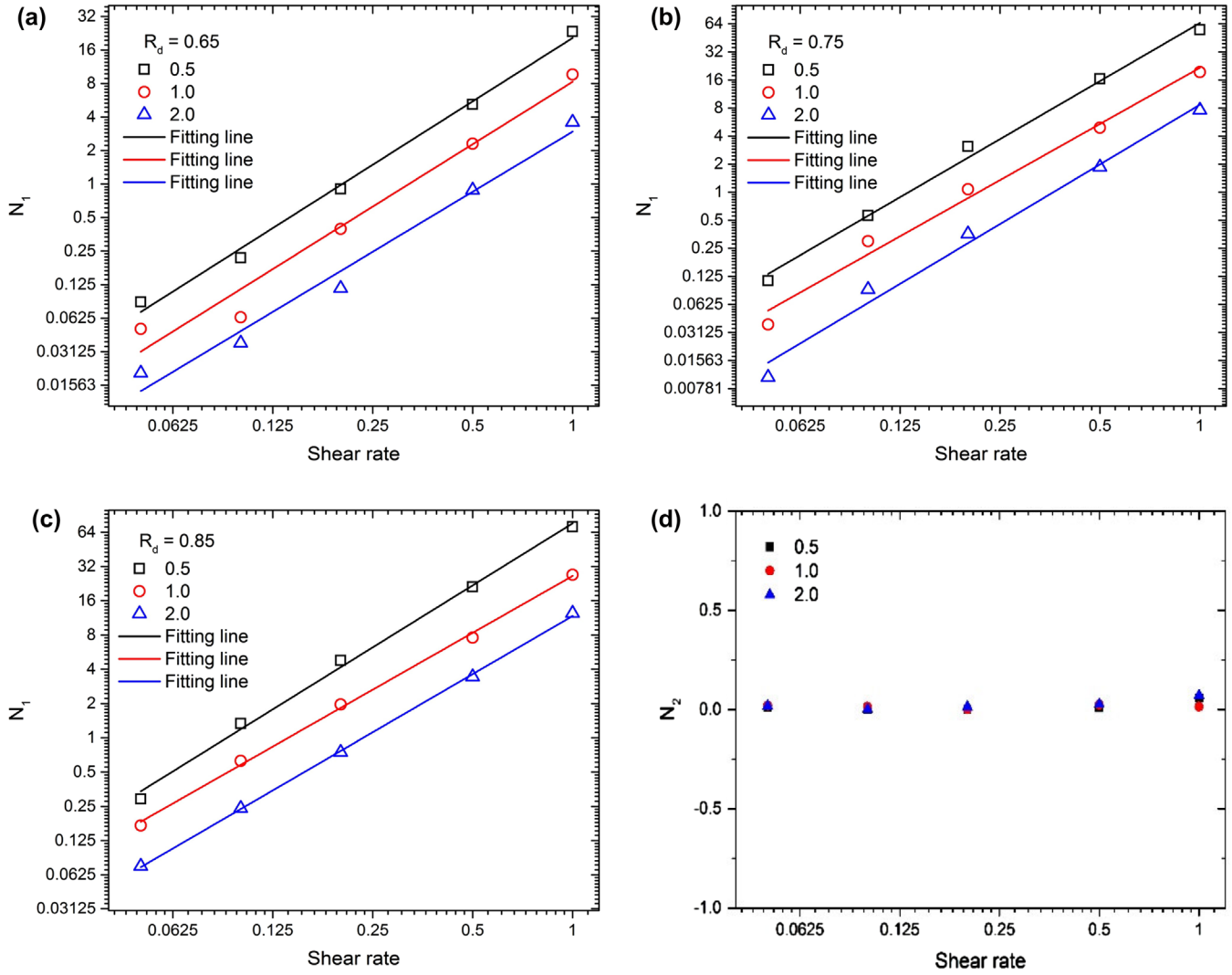


Figure 16. (Colour online) Relation between N_1 and the shear rate, ranging from 0.05 to 1.0, and spring constant fixed at 0.5, 1.0, 2.0, (a) $r_d = 0.65$, (b) $r_d = 0.75$, (c) $r_d = 0.85$, (d) relation between N_2 and the shear rate.

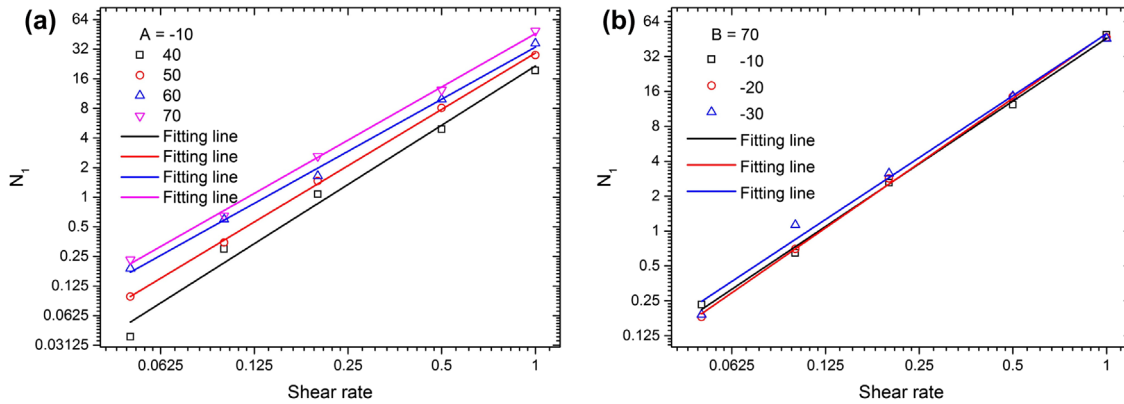


Figure 17. (Colour online) Relation between N_1 and the shear rate at fixed coefficients, (a) $A = -10$, (b) $B = 70$.

Here, S_C and S_D are the contribution to the stress by the conservative and dissipative forces, respectively. Similarly, the viscosity is divided into the conservative η_C and dissipative η_D part, calculating from S_C and S_D . From Figure 10, we find that the dissipative viscosity η_D matches well with Marsh's prediction [9].

However, the conservative viscosity η_C cannot be neglected any more compared with η_D , since the potential interactions in MDPD is much higher than that in DPD. In a previous paper, Jamali et al. [41] pointed out that increasing repulsive coefficient directly increased the viscosity for a MDPD fluid. Due to the

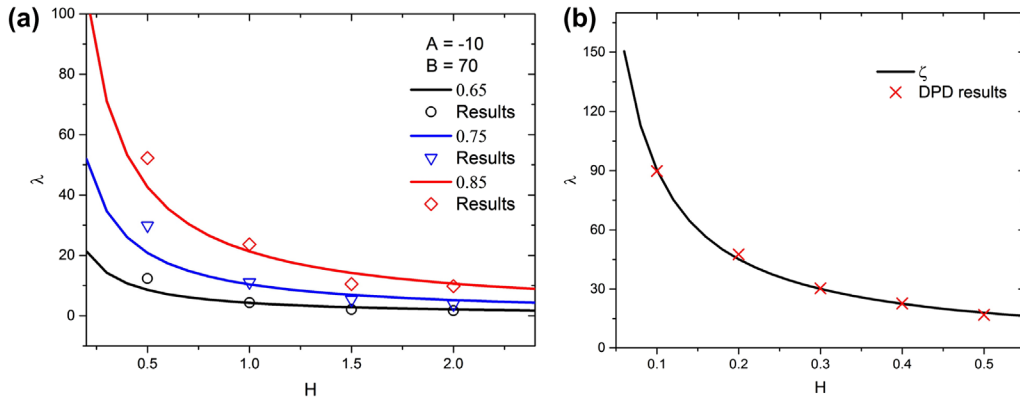


Figure 18. (Colour online) Variation of relaxation time versus spring constant (a) $A = -10$, $B = 70$, and $r_d = 0.65, 0.75, 0.85$ for MDPD model, (b) DPD model.

presence of attractive forces in MDPD, the MDPD fluid tends to retract and reduce its volume at a low pressure. In our simulation, the calculations with unphysical negative pressure are discarded and only results satisfying the canonical NVT ensemble are retained.

In Figure 11, the curves increasing with the repulsive coefficient represent the conservative viscosity with different values of the attractive coefficient, and the ones that are relatively constant represent the dissipative viscosity. Figure 11 reveals that the attractive and repulsive coefficients are not a function of dissipative viscosity η_D ; in contrast, η_C sharply increases with increasing repulsive coefficient, consistent with Jamali et al.'s conclusion [41]. Furthermore, η_C is affected by the attractive coefficient as well, contrary to Jamali et al. [41]. Figure 12 shows that the ratio of η_C to η_D obviously increases with high density where the effect of many-body interactions is significant.

The comparison between the conservative viscosity η_C and the dissipative viscosity η_D under various densities at two cut-off radius values $r_d = 0.65$ and 0.85 is similar to that one at $r_d = 0.75$. η_C apparently increases with increasing repulsive coefficient and increasing density. At low density, cut-off radius and repulsive coefficient, η_C is smaller than η_D , i.e. η_D dominates MDPD viscosity. Generally, η_C becomes comparable to η_D with increasing density and repulsive coefficient. This is the reason why Marsh concludes that his prediction is merely appropriate for dilute gases and fluids system, i.e. at low density which is not feasible for MDPD systems. MDPD aims to simulate dense liquids with sharp interface and strong many-body interactions. Furthermore, it is noted that the effect of the attractive coefficient on η_C is non-negligible at low r_d , yet, this effect vanishes at high r_d . In that situation, many-body terms of repulsion completely dominate the system, leading to a considerably high pressure. In contrast to attractive forces which are mainly due to two-body are not further able to obviously influence the pressure. Hence, Jamali et al. [41] concluded that viscosity is not affected by the attractive force, because his simulations focus on high density using Green-Kubo equation. The ratio of η_C to η_D for different values of density is shown in Figure 13, revealing ρ and r_d significant affect η_C . Figure 14 shows that the logarithmic slope of η_C versus the density is not apparently the same at different cut-off radius r_d .

The range of slopes for all cases is approximately 4.0–5.0, with some deviations at low r_d and large repulsive coefficient.

According to Marsh's theoretical prediction [9], the viscosity depends on the quadratic term in the density due to the two-body interactions for standard DPD model. Here the higher order interactions dominate MDPD system. Additionally, we also find that the viscosity is basically linear to quadratic in the repulsive coefficient. Thus, we suggest a simple fitting equation for predicting the conservative viscosity η_C

$$\eta_C \cong k\rho^\alpha B^2, \quad (20)$$

where k and α are two fitting parameters, similar to c in Equation (18), depending on the cut-off radius r_d . These parameters are listed in Table 3.

Figure 15 shows that the Equation (20) is able to predict the conservative viscosity reasonably well, especially at a high cut-off radius, where the many-body interactions dominate. In summary, we arrive at the following viscosity prediction for MDPD system ($s = 0.5$):

$$\eta = \eta_C + \eta_D \cong k\rho^\alpha B^2 + \frac{512\pi\gamma\rho^2 r_c^5}{51,975} \quad (21)$$

3.4. Dumbbell polymer model for MDPD

In this section, we establish an MDPD model for a suspension of dumbbells in a MDPD fluid, and simulate its simple shear flow with Lees-Edward boundary condition. The two MDPD particles, treated as beads in polymers, are connected by spring force as suggested by Equation (10). In implementing the periodic boundary condition, the positions of dumbbell beads are also independently recorded. The volume fraction (number fraction, these particles has the same mass) of dumbbells is fixed at 5.88%, the density at 4.25 and the shear rate is varied ranging from 0.05 to 1.0. Firstly, Figure 15 indicates that the relation between the positive first normal stress difference N_1 and the shear rate is consistent with the theoretical consideration, Equation (15). The slopes are found in the range from 1.8 to 2.1 with zero second normal stress difference N_2 , in agreement to Pan et al.'s work [50], regardless of the cut-off radius r_d (Figure 16(a)–(d)).

Next, we focus on the influence of the attractive and repulsive coefficients on the first normal stress difference. In Figure 17(a), the attractive coefficient is fixed at -10 with increasing repulsive coefficient from 40 to 70. It reveals that N_1 increases with increasing repulsive coefficient. In contrast, Figure 17(b)

suggests that N_1 is not greatly affected by the attractive coefficient, since the attractive coefficient does not evidently influence the viscosity, which we have mentioned in the last section.

Finally, on the basis of Equation (15), the relaxation time λ can be derived from N_1 . The relationship between λ and spring constant H can be shown to obey Equation (16). Figure 18(a) shows that λ matches well with increasing H in MDPD model, irrespectively of r_d . However, the DPD results agree much better with the theory, as shown in Figure 18(b), which may be due to the lack of strong many-body interactions in the standard DPD model.

4. Conclusions

The viscosity and a dumbbell polymeric suspension model are developed and calculated for the many-body dissipative particle dynamics (MDPD). First, we find that the EOS of MDPD, established by Jamali et al. [41], is limited to a special cut-off radius r_d ($= 0.75$). It needs to be modified through allowing the specific density correction c to depend on the cut-off radius r_d . Next, after validating the Lees–Edward boundary condition under high dissipative parameter γ , we investigate the viscosity of the MDPD system, especially focusing on the conservative viscosity η_C deriving from the conservative force. Due to the strong many-body interactions, η_C cannot be neglected in the MDPD model as compared to the standard DPD model. Our results suggest that the repulsive coefficient, the density ρ , the cut-off radius r_d significantly influence η_C . η_C monotonically increases with these parameters. The effect of the attractive coefficient on η_C is not significant, however. On the basis of our numerical results, a prediction for the fluid viscosity is proposed which works well at high r_d . Finally, the dumbbell polymeric suspension model is established via the MDPD system. The first normal stress difference N_1 is as a function of shear rate consistent with the theoretical prediction of Oldroyd-B model, and the second normal stress difference N_2 is zero in the mean. Increasing the repulsive coefficient increases N_1 ; the attractive coefficient does not seem to affect N_1 . These simulations pave the way for MDPD as a model for non-Newtonian droplets.

Disclosure statement

No potential conflict of interest was reported by the authors.

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References

- [1] Benzi R, Succi S, Vergassola M. The lattice Boltzmann equation: theory and applications. *Phys Rep*. 1992;222:145–197.
- [2] Chen S, Doolen GD. Lattice Boltzmann method for fluid flows. *Annu Rev Phys Chem*. 1998;30:329–364.
- [3] Jacquin H, Kim B, Kawasaki K, et al. Brownian dynamics: from glassy to trivial. *Phys Rev E*. 2015;91:022130.
- [4] Hoogerbrugge PJ, Koelman JMVA. Simulating microscopic hydrodynamic phenomena with dissipative particle dynamics. *Europhys Lett (EPL)*. 1992;19:155.
- [5] Español P. Hydrodynamics from dissipative particle dynamics. *Phys Rev E*. 1995;52:1734.
- [6] Groot RD, Warren PB. Dissipative particle dynamics: bridging the gap between atomistic and mesoscopic simulation. *J Chem Phys*. 1997;107(11):4423–4435.
- [7] Español P, Warren PB. Statistical mechanics of dissipative particle dynamics. *Europhys Lett*. 1995;30(4):191.
- [8] Marsh CA, Backx G, Ernst MH. Fokker–Planck–Boltzmann equation for dissipative particle dynamics. *Europhys Lett*. 1997;38(6):411.
- [9] Marsh CA, Backx G, Ernst MH. Static and dynamic properties of dissipative particle dynamics. *Phys Rev E*. 1997;56:1676.
- [10] Satoh A, Majima T. Comparison between theoretical values and simulation results of viscosity for the dissipative particle dynamics method. *J Colloid Interface Sci*. 2005;283(1):251–266.
- [11] Fedosov DA, Caswell B, Karniadakis GE. A multiscale red blood cell model with accurate mechanics, rheology, and dynamics. *Biophys J*. 2010;98:2215–2225.
- [12] Fan X, Phan-Thien N, Chen S, et al. Simulating flow of DNA suspension using dissipative particle dynamics. *Phys Fluids*. 2006;18:063102.
- [13] Spenley NA. Scaling laws for polymers in dissipative particle dynamics. *Europhys Lett*. 2000;49(4):534.
- [14] Chen S, Phan-Thien N, Fan XJ, et al. Dissipative particle dynamics simulation of polymer drops in a periodic shear flow. *J Nonnewton Fluid Mech*. 2004;118(1):65–81.
- [15] Pagonabarraga I, Frenkel D. Dissipative particle dynamics for interacting systems. *J Chem Phys*. 2001;115(11):5015–5026.
- [16] Trofimov SY, Nies ELF, Michels MAJ. Thermodynamic consistency in dissipative particle dynamics simulations of strongly nonideal liquids and liquid mixtures. *J Chem Phys*. 2002;117(20):9383–9394.
- [17] Pagonabarraga I, Frenkel D. Non-ideal DPD fluids. *Mol Simul*. 2000;25:3:167–175.
- [18] Warren PB. Hydrodynamic bubble coarsening in off-critical vapor–liquid phase separation. *Phys Rev Lett*. 2001;87:225702.
- [19] Warren PB. Vapour–liquid coexistence in many-body dissipative particle dynamics. *Phys Rev E*. 2003;68:066702.
- [20] Ghoufi A, Malfreyt P. Calculation of the surface tension from multibody dissipative particle dynamics and Monte Carlo methods. *Phys Rev E*. 2010;82:016706.
- [21] Ghoufi A, Malfreyt P. Mesoscale modeling of the water liquid–vapor interface: a surface tension calculation. *Phys Rev E*. 2011;83:051601.
- [22] Chen C, Gao C, Zhuang L, et al. A many-body dissipative particle dynamics study of spontaneous capillary imbibition and drainage. *Langmuir*. 2010;26(12):9533–9538.
- [23] Zhao JY, Chen S, Liu Y. Droplets motion on chemically/topographically heterogeneous surfaces. *Mol Simul*. 2016;42(17):1452–1459.
- [24] Zhao JY, Chen S, Liu Y. Dynamical behaviors of droplet impingement and spreading on chemically heterogeneous surfaces. *Appl Surf Sci*. 2017;400:1.
- [25] Wang YX, Chen S. Numerical study on droplet sliding across micropillars. *Langmuir*. 2015;31(16):4673–4677.
- [26] Boromand A, Jamali S, Maia JM. Viscosity measurement techniques in dissipative particle dynamics. *Comput Phys Commun*. 2015;196:1.
- [27] Meier K, Laesecke A, Kabelac S. Transport coefficients of the Lennard–Jones model fluid. I. Viscosity. *J Chem Phys*. 2004;121(8):3671–3687.
- [28] Green MS. Markoff random processes and the statistical mechanics of time-dependent phenomena. *J Chem Phys*. 1952;20(8):1281–1295.
- [29] Kubo R. Statistical-mechanical theory of irreversible processes. I. *J Phys Soc Jpn*. 1957;12(6):570–586.
- [30] Arya G, Maginn EJ, Chang HC. Efficient viscosity estimation from molecular dynamics simulation via momentum impulse relaxation. *J Chem Phys*. 2000;113:2079.
- [31] Backer JA, Lowe CP, Hoefsloot HCJ, et al. Poiseuille flow to measure the viscosity of particle model fluids. *J Chem Phys*. 2005;122:154503.

- [32] Fedosov DA, Karniadakis GE, Caswell B. Steady shear rheometry of dissipative particle dynamics models of polymer fluids in reverse Poiseuille flow. *J Chem Phys.* **2010**;132:144103.
- [33] Chen S, Phan-Thien N, Khoo BC, et al. Flow around spheres by dissipative particle dynamics. *Phys Fluids.* **2006**;18(10):103605.
- [34] Moshfegh A, Ahmadi G, Jabbarzadeh A. Thermostatic and rheological responses of DPD fluid to extreme shear under modified Lees–Edwards boundary condition. *EurPhys J E.* **2015**;38:134.
- [35] Moshfegh A, Jabbarzadeh A. Modified Lees–Edwards boundary condition for dissipative particle dynamics: hydrodynamics and temperature at high shear rates. *Mol Simul.* **2015**;41(15):1264–1277.
- [36] Evans GT. Dissipative particle dynamics: transport coefficients. *J Chem Phys.* **1999**;110(3):1338–1342.
- [37] Masters AJ, Warren PB. Kinetic theory for dissipative particle dynamics: the importance of collisions. *Europhys Lett (EPL).* **1999**;48:1.
- [38] Chaudhri A, Lukes JR. Velocity and stress autocorrelation decay in isothermal dissipative particle dynamics. *Phys Rev E.* **2010**;81:026707.
- [39] Visser DC, Hoefsloot HCJ, Iedema PD. Modelling multi-viscosity systems with dissipative particle dynamics. *J Comput Phys.* **2006**;214(2):491–504.
- [40] Jung G, Schmid F. Computing bulk and shear viscosities from simulations of fluids with dissipative and stochastic interactions. *J Chem Phys.* **2016**;144:204104.
- [41] Jamali S, Boromand A, Khani S, et al. Generalized mapping of multi-body dissipative particle dynamics onto fluid compressibility and the Flory–Huggins theory. *J Chem Phys.* **2015**;142:164902.
- [42] Yamada T, Yuan J, Sunden BÅ. Application of many-body dissipative particle dynamics to determine liquid characteristics. *Int J Numer Methods Heat Fluid Flow.* **2015**;25(7):1619–1637.
- [43] Chatterjee A. Modification to Lees–Edwards periodic boundary condition for dissipative particle dynamics simulation with high dissipation rates. *Mol Simul.* **2007**;33(15):1233–1236.
- [44] Lowe C. An alternative approach to dissipative particle dynamics. *Europhys Lett (EPL).* **1999**;47:145.
- [45] Stoyanov SD, Groot RD. From molecular dynamics to hydrodynamics: a novel Galilean invariant thermostat. *J Chem Phys.* **2005**;122:114112.
- [46] Arienti M, Pan W, Li X, et al. Many-body dissipative particle dynamics simulation of liquid/vapor and liquid/solid interactions. *J Chem Phys.* **2011**;134:204114.
- [47] Ghoufi A, Malfreyt P. Coarse grained simulations of the electrolytes at the water–air interface from many body dissipative particle dynamics. *J Chem Theory Comput.* **2012**;8(3):787–791.
- [48] Lees AW, Edwards F. The computer study of transport process under extreme conditions. *J Phys C.* **1972**;5:1921.
- [49] Chen S, Jin Y, Zhang M, et al. Imposing Lees–Edwards boundary conditions in dissipative particle dynamics. *J Tongji Univ(Nat Sci).* **2012**;40:137–142.
- [50] Pan D, Hu J, Shao X. Lees–Edwards boundary condition for simulation of polymer suspension with dissipative particle dynamics method. *Mol Simul.* **2016**;42(4):328–336.
- [51] Phan-Thien N. Understanding viscoelasticity: an introduction to rheology, graduate texts in physics. Berlin: Springer-Verlag; **2013**. ISBN 978-3-642-32957-9.
- [52] Kirkwood JG, Buff FP. The statistical mechanical theory of surface tension. *J Chem Phys.* **1949**;17(3):338–343.
- [53] Irving JH, Kirkwood JG. The statistical mechanical theory of transport processes. IV. The equations of hydrodynamics. *J Chem Phys.* **1950**;18(6):817–829.
- [54] Enrique DM, Guy G, Felipe B, et al. Test-area simulation method for the direct determination of the interfacial tension of systems with continuous or discontinuous potentials. *J Chem Phys.* **2005**;123(13):134703.
- [55] Mo CJ, Yang LJ, Zhao F, et al. Mesoscopic simulation of a thinning liquid bridge using the dissipative particle dynamics method. *Phys Rev E.* **2015**;92:023008.
- [56] Pagonabarraga I, Hagen MHJ, Frenkel D. Self-consistent dissipative particle dynamics algorithm. *Europhys Lett.* **1998**;42(4):377.
- [57] Masters AJ, Warren PB. Kinetic theory for dissipative particle dynamics: the importance of collisions. *Europhys Lett (EPL).* **1999**;48:1.
- [58] Noguchi H, Gompper G. Transport coefficients of dissipative particle dynamics with finite time step. *Europhys Lett (EPL).* **2007**;79:36002.
- [59] Fan X, Phan-Thien N, Teng YN, et al. Microchannel flow of a macromolecular suspension. *Phys Fluids.* **2003**;15(1):11–21.
- [60] Chen S, Phan-Thien N, Khoo BC, et al. Flow around spheres by dissipative particle dynamics. *Phys Fluids.* **2006**;18:10.